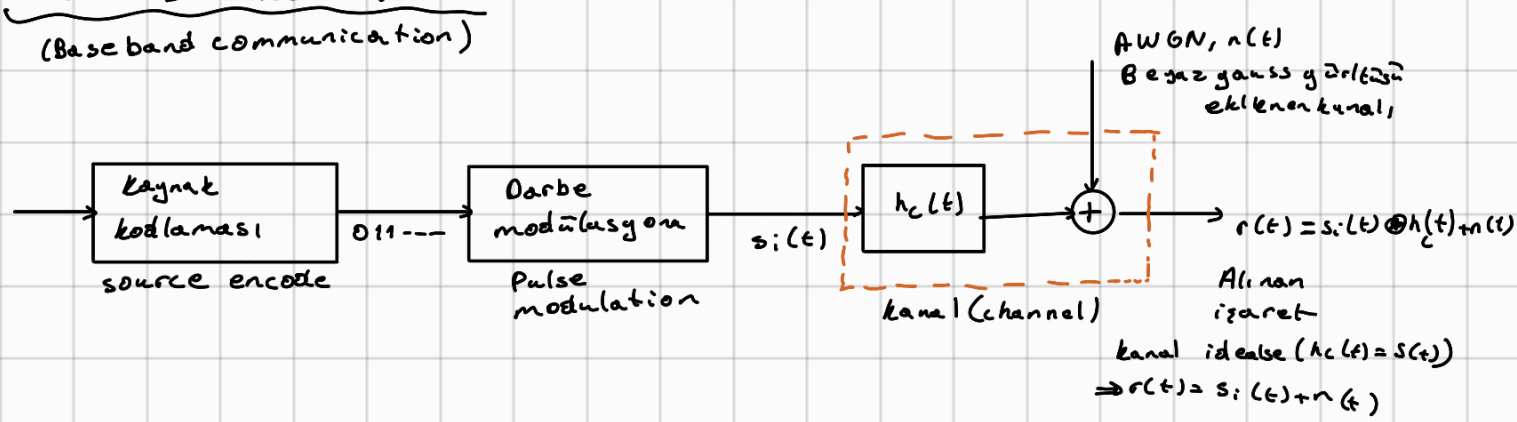
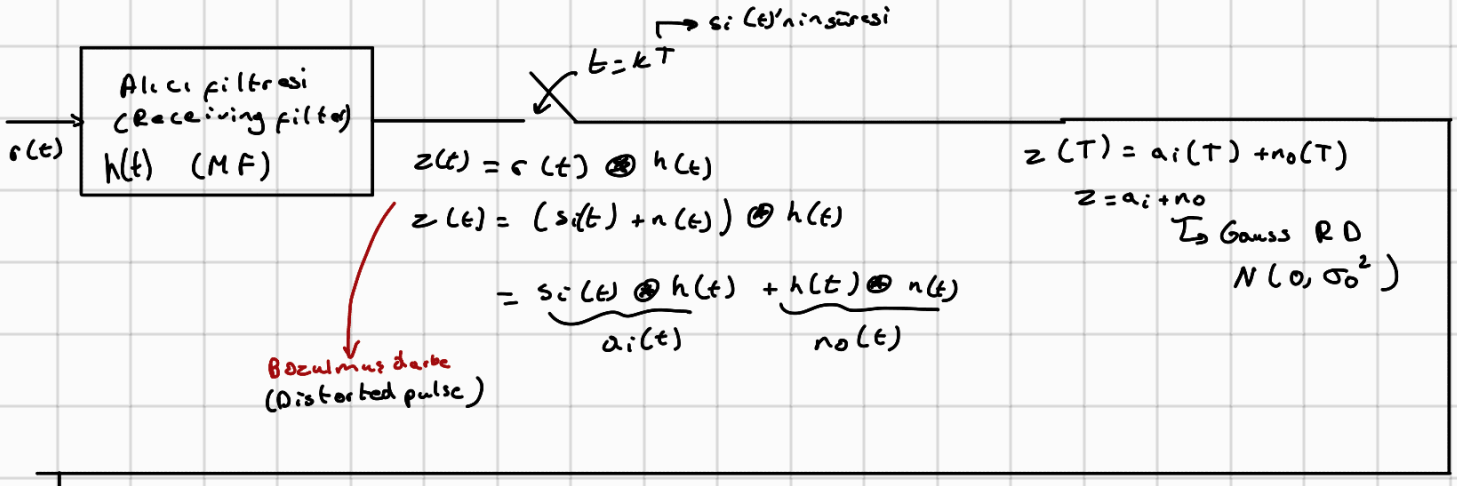
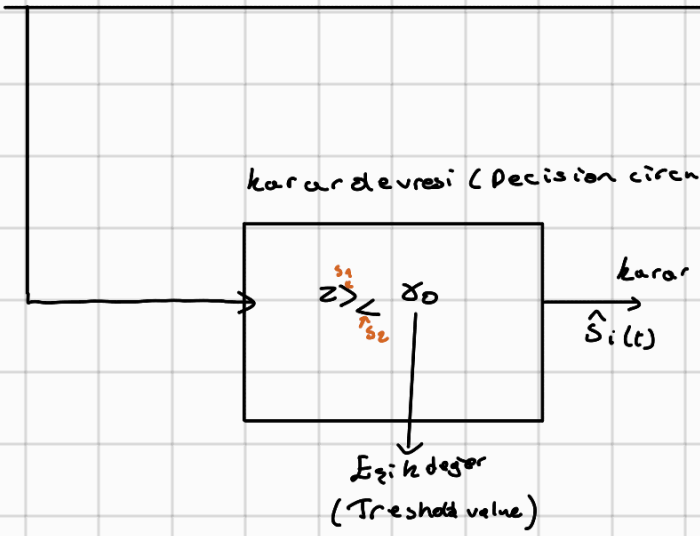
Ders 8

Temel Band haberleşmesi
(Baseband communication)







$$z = a_i + n_0 \qquad \hookrightarrow \sim N(0, \sigma_0^2)$$

$$\downarrow$$

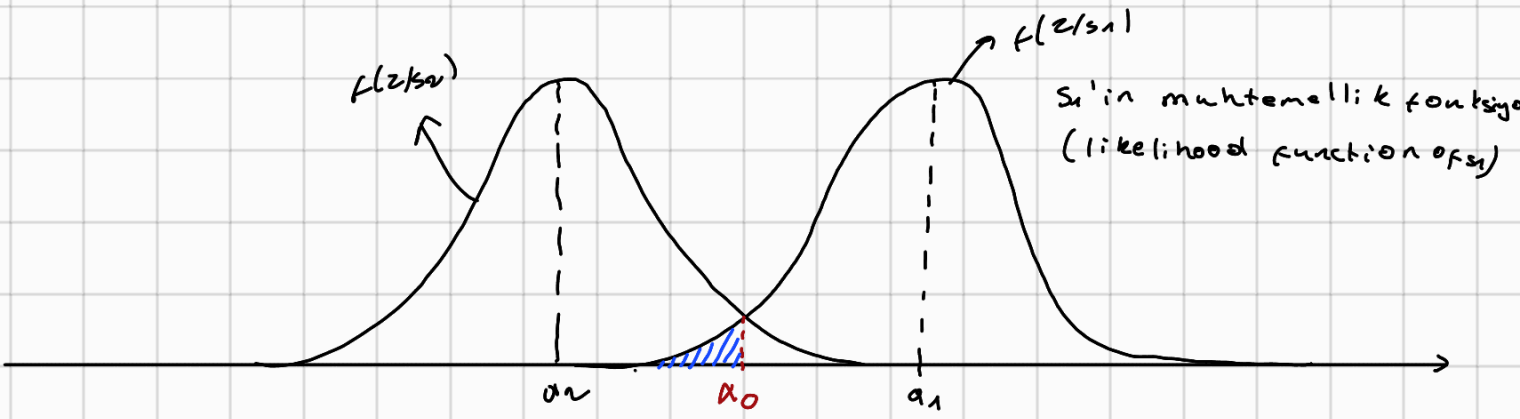
$$\sim N(a_i, \sigma_0^2)$$

$$f(n_0) = \frac{1}{\sqrt{2\pi}\,\sigma_0} \exp\left\{-\frac{1}{2}\left(\frac{n_0}{\sigma_0}\right)^2\right\}$$

Eğer s_1 iletilirse $\quad z = a_1 + n_0 \quad \hookrightarrow N(a_1, \sigma_0^2) \quad \Rightarrow f(z/s_1) = \frac{1}{\sqrt{2\pi}\,\sigma_0} \exp\left\{-\frac{(z-a_1)^2}{2\sigma_0^2}\right\}$

Eğer s_2 iletilirse $\quad z = a_2 + n_0 \quad \Rightarrow f(z/s_2) = \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{(z-a_2)^2}{2\sigma_0^2}\right\}$

$\hookrightarrow N(a_2, \sigma_0^2)$



$s_1(t)$ iletilmiş olsun : Doğru karar ($z > \gamma_0$)

Hata : $P(h/s_1) = P(z < a_0) = \int_{-\infty}^{\gamma_0} f(z/s_1)\,dz$

$s_2(t)$ iletilmiş olsun doğru karar: ($z < \gamma_0$)

Hata : $P(h/s_2) = P(z > \gamma_0) = \int_{\gamma_0}^{\infty} f(z/s_2)\,dz$

$$P_b = \sum_{i=1}^{2} P(h/s_i)P(s_i) = P(h/s_1)P(s_1) + P(h/s_2)P(s_2)$$

P_b ← Bit hata olasılığı

Eşit olasılıklı işaretler: (Equality likely) $\quad P(s_1) = P(s_2) = \frac{1}{2}$

$\Rightarrow \gamma_0 = \frac{a_1 + a_2}{2} \quad \Rightarrow$ Simetriden dolayı

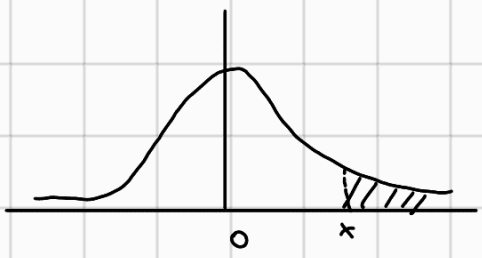
$P(h/s_1) = P(h/s_2)$

$\Rightarrow P_b = P(h/s_1) = P(h/s_2)$

$$P_b = P(h/s_2) = \int_{\gamma_0}^{\infty} f(z/s_2) = \frac{1}{\sqrt{2\pi}\,\sigma_0} \int_{\frac{a_1+a_2}{2}}^{\infty} e^{-\frac{(z-a_2)^2}{2\sigma_0^2}}\,dz$$

$Q(x) = P(X > x)$

$X \sim N(0,1)$



$$Q(x) = P(X > x) = \int_x^\infty f_X(x)\,dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-\frac{t^2}{2}}\,dt, \quad x \geq 0$$

Table B.1, Page 104s

$Q(0) = 0,5$

$Q(-x) = 1 - Q(x)$

$$P_0 = P(h/s_2) = \int_{\sigma_0}^\infty f(z/s_2) = \frac{1}{\sqrt{2\pi}\,\sigma_0} \int_{\frac{a_1+a_2}{2}}^\infty e^{-\frac{(z-a_2)^2}{2\sigma_0^2}}\,dz = \frac{1}{\sqrt{2\pi}\,\sigma_0} \int_{\frac{a_1-a_2}{2\sigma_0}}^\infty e^{-\frac{t^2}{2}}\sigma_0\,dt = Q\left(\frac{a_1-a_2}{2\sigma_0}\right)$$

$\frac{z-a_2}{\sigma_0} = t$

$z = \sigma_0 t + a_2$

$dz = \sigma_0\,dt$

$z = \frac{a_1+a_2}{2} \rightarrow t = \frac{\frac{a_1+a_2}{2} - a_2}{\sigma_0} = \frac{a_1-a_2}{2\sigma_0}$

$z = \infty \rightarrow t = \infty$

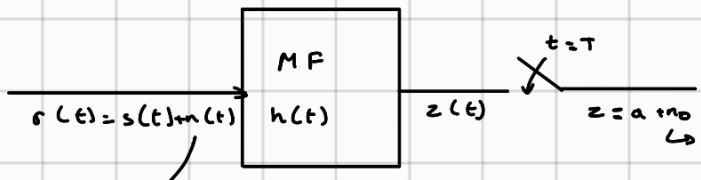
$$P_0 = Q\left(\frac{a_1-a_2}{2\sigma_0}\right)$$

burası ne kadar artarsa o kadar P_0 azalır

Karşılaştırıcı Filtre

(Matched Filter - MF)

- MF provides max SNR at its output for a given transmitted waveform



$S_n(t) = \frac{N_0}{2}$ $\frac{Watt}{hz}$

$$SNR = \frac{a^2}{\sigma_0^2}$$

a^2 → ani işaret gücü (instantaneous signal power)

σ_0^2 → ortalama gürültü gücü (average noise power)

$$SNR = \left(\frac{S}{N}\right)_{t=T} = \frac{a^2}{\sigma_0^2} = \frac{2E_s}{N_0}$$

$\frac{N_0}{2}$ ←

$$E_s = \int_0^T |s(t)|^2\,dt$$

eğer $h(t) = s(T-t)$, $0 \leq t \leq T$

$$z(t) = r(t) \otimes h(t) = \underbrace{s(t) \otimes h(t)}_{a(t)} + \underbrace{n(t) \otimes h(t)}_{n_0(t)}$$

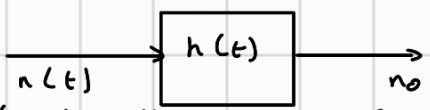
$$z(T) = a(T) + n_0(T)$$

$$z = a + n_0 \hookrightarrow N(0, \sigma_0^2)$$

$$SNR = \left(\frac{S}{N}\right)_T = \frac{a^2}{\sigma_0^2}$$

$$a(t) = F^{-1}\{A(f)\} = \int_{-\infty}^{\infty} S(f)\, H(f)\, e^{j2\pi fT} df$$

$$\longrightarrow a = a(T)$$



$S_n(f) = \frac{N_0}{2}$ watt/Hz (GYF)

$$S_{n0}(f) = S_n(f)\,|H(f)|^2 = \frac{N_0}{2}\,|H(f)|^2$$

$$P_{n_0} = \sigma_0^2 = \int_{-\infty}^{\infty} S_{n0}(f)\, df = \frac{N_0}{2} \int_{-\infty}^{\infty} |H(f)|^2\, df$$

$$SNR = \frac{a^2}{\sigma_0^2} = \frac{\left| \int_{-\infty}^{+\infty} S(f) H(f) e^{j2\pi fT} df \right|^2}{\frac{N_0}{2} \int_{-\infty}^{\infty} |H(f)|^2 df}$$

Cauchy-Schwarz eşitsizliği:

$$\left| \int_{-\infty}^{\infty} f_1(x) f_2(x)\, dx \right|^2 \le \int_{-\infty}^{\infty} |f_1(x)|^2 dx \int_{-\infty}^{\infty} |f_2(x)|^2 dx$$

$\hookrightarrow f_1(x) = k \cdot f_2^*(x)$ ise eşittir

$f_1(x) = H(f)$

$f_2(x) = S(f) e^{j2\pi fT}$

$$\Rightarrow SNR \le \frac{\int_{-\infty}^{\infty} |H(f)|^2 df \int_{-\infty}^{\infty} |S(f)|^2 df}{\frac{N_0}{2} \int_{-\infty}^{\infty} |H(f)|^2 df}$$

$$SNR \le \frac{2}{N_0} \underbrace{\int_{-\infty}^{\infty} |S(f)|^2 df}_{E_s} \Rightarrow SNR \le \frac{2E_s}{N_0}$$

SNR'in maksimum değeri budur

Biz mak bulmak istiyoruz → $SNR = \frac{2E_s}{N_0}$ olması için

$$f_1(x) = k_2 f_2^*(x)$$

$$H(f) = k\, S^*(f)\, e^{-j2\pi fT}$$

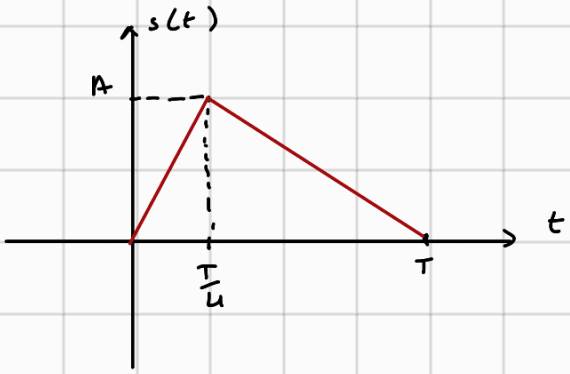
$$\downarrow$$

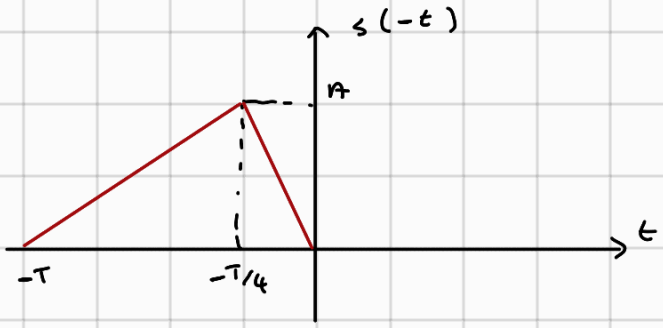
$$h(t) = k \int_{-\infty}^{\infty} \underbrace{S^*(f)\, e^{-j2\pi fT}}\, e^{j2\pi ft}\, df$$

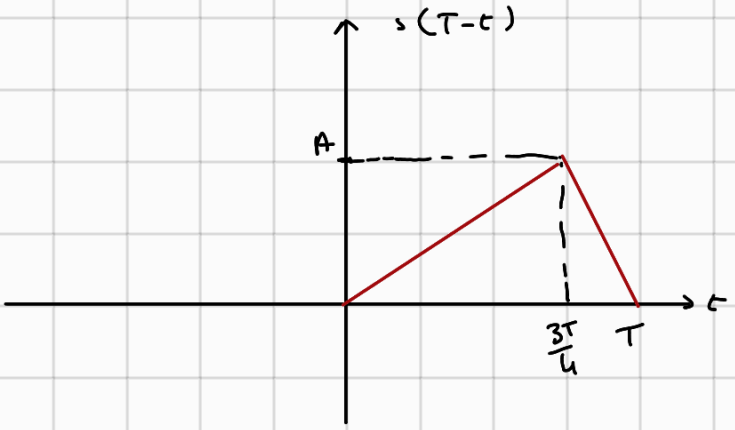
$$= k \underbrace{\left[\int_{-\infty}^{\infty} s(t)\, e^{j2\pi f(\underbrace{T-t}_{t'})}\, dt \right]}_{s(t')}$$

$$h(t) = k \cdot s^*(t') = k\, s^*(T-t) \rightarrow \boxed{h(t) = s^*(T-t)}$$

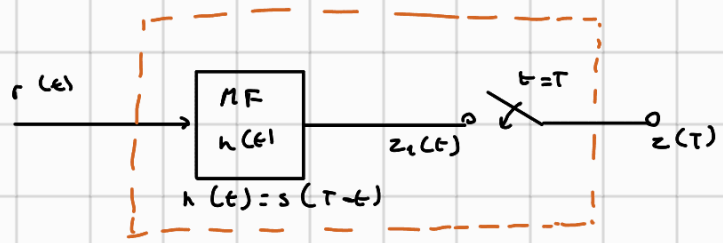
$h(t) = s(T-t) \Rightarrow SNR = \frac{2E_s}{N_0}$



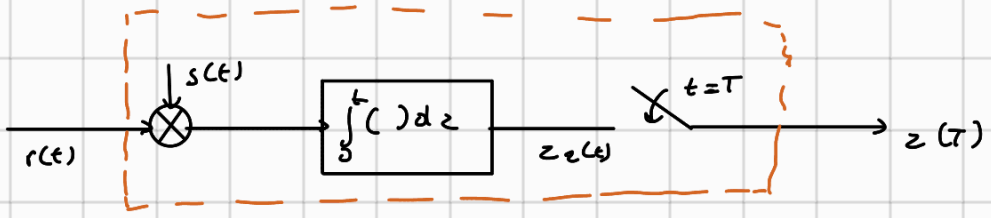




İlişkilendirici (Correalator)



≡



$$z_1(t) = r(t) \circledast h(t) = \int_0^t r(z)\, h(t-z)\, dz$$

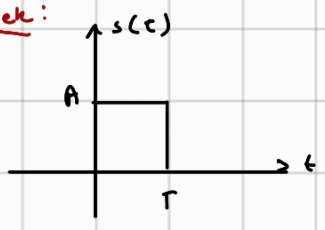
$$z_1(t) = \int_0^t r(z)\, s(T-(t-z))\, dz$$

$$z_1(T) = \int_0^T r(z)\, s(z)\, dz = z(T)$$

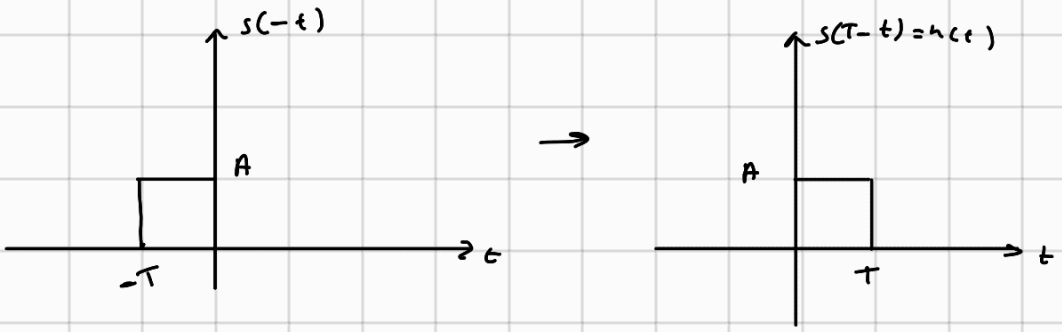
$$z_2(t) = \int_0^t r(z)\, s(z)\, dz$$

$$z_2(T) = \int_0^T r(z)\, s(z)\, dz = z(T)$$

Örnek:

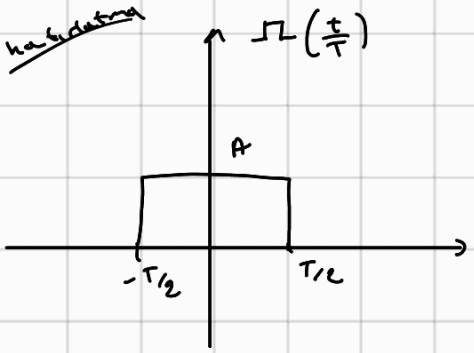


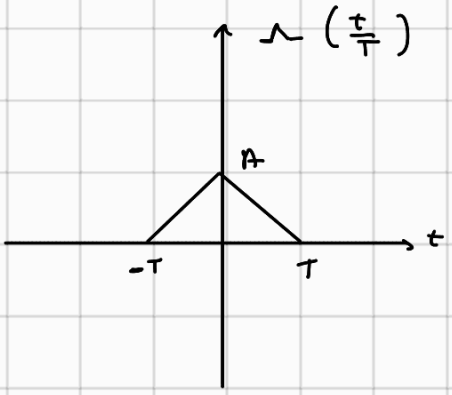
$r(t) = s(t)$ (no noise)



MF: $z_1(t) = r(t) \otimes h(t) = s(t) \otimes s(t)$

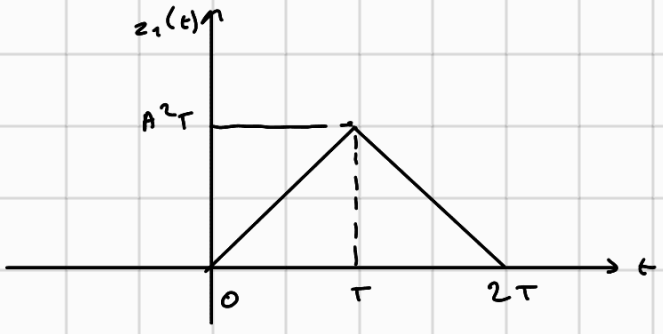
A hatırlatma





$$A^2 T \, \Lambda\left(\frac{t}{T}\right) = \Pi\left(\frac{t}{T}\right) \otimes \Pi\left(\frac{t}{T}\right)$$

$\Rightarrow z_1(t)$



İlişkilendirici

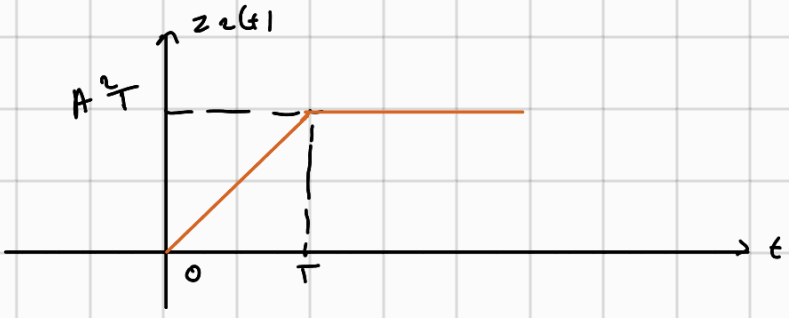
$$z_2(t) = \int_0^t r(z)\, s(z)\, dz = \int_0^t s^2(z)\, dz \qquad = 0, \quad t < 0$$

$r(z) = s(z)$

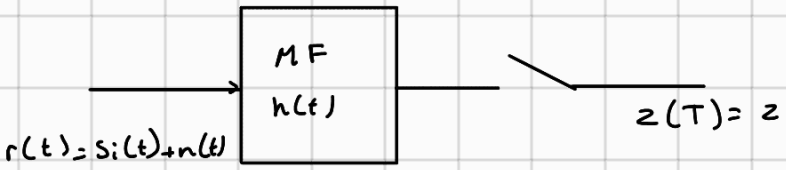
$0 < t < T$

$$= \int_0^t A^2 dz = A^2 t, \quad 0 < t < T$$

$$= \int_0^T A^2 dz = A^2 T, \quad t > T$$

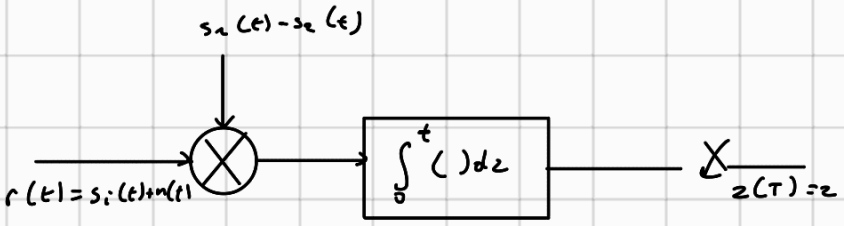


$$z_2(T) = z(T) = A^2 T$$



matched to $s_1(t) \rightarrow z_2(t)$

$$h(t) = s_1(T-t) - s_2(T-t)$$



$$P_b = Q\left(\frac{a_1 - a_2}{2\sigma_0}\right) \qquad \left(\frac{a_1 - a_2}{\sigma_0}\right)^2 = \frac{(a_1 - a_2)^2}{\sigma_0^2}$$

$$\left(\frac{S}{N}\right)_T = \frac{a^2}{\sigma_0^2} = \frac{2E_b}{N_0}$$ ← s(t)'nin enerjisi ← h(t) ile s(t)'yi karşılaştığımızda

$$\left(\frac{S}{N}\right)_T = \frac{(a_1 - a_2)^2}{\sigma_0^2} = \frac{2E_d}{N_0}$$ ← Fark işaretin enerjisi (Difference signal) ← h(t) $s_1(t) - s_2(t)$'yi karşılaşınca

$$E_d = \int_0^T [s_1(t) - s_2(t)]^2 dt = \underbrace{\int_0^T s_1^2(t)\,dt}_{E_b} + \underbrace{\int_0^T s_2^2(t)\,dt}_{E_b} - 2\underbrace{\int_0^T s_1(t)\,s_2(t)\,dt}_{\rho \cdot E_b}$$

E_b : bit başına enerji per bit

$$\rho = \frac{1}{E_b}\int_0^T s_1(t)\,s_2(t)\,dt$$

↓ karşı - ilişki katsayısı

(Normalized cross corelation coefficient)

$$-1 \le \rho \le 1$$

$$E_d = 2E_b - 2\rho E_b = 2E_b(1-\rho)$$

$$P_b = Q\left(\frac{a_1 - a_2}{2\sigma_0}\right) = Q\left(\sqrt{\frac{2E_d}{4N_0}}\right) = Q\left(\sqrt{\frac{E_d}{2N_0}}\right) = Q\left(\sqrt{\frac{E_b}{N_0}(1-\rho)}\right)$$